ABES Engineering College, Ghaziabad

Department of Applied Sciences & Humanities

Session: 2025-26 **Semester: I**

Course Code: 25AS102

Course Name: Fundamentals of Linera Algebra & Statistics **Max. Marks: 10**

Assignment 1

Date of Assignment: 22/09/25 **Date of submission:24/09/25**

S.No.	KL	CO	Question	Marks
1	K3	CO3	Employing elementary row transformations, find the inverse of the matrix $A = \begin{bmatrix} 0 & 1 & 2 \\ 1 & 2 & 3 \\ 3 & 1 & 1 \end{bmatrix}$.	2
2.	K3	CO3	Reduce the matrix A to its normal form and hence find its rank where, $A= \begin{bmatrix} 2 & 1 & -3 & -6 \\ 3 & -3 & 1 & 2 \\ 1 & 1 & 1 & 2 \end{bmatrix}$	2
3.	K3	CO3	Find the Eigen values and Eigen vectors of the following matrices: $A = \begin{bmatrix} 6 & -2 & 2 \\ -2 & 3 & -1 \\ 2 & -1 & 3 \end{bmatrix}$	2
4	K3	CO3	Verify Cayley-Hamilton theorem for the matrix $A = \begin{bmatrix} 2 & -1 & 1 \\ -1 & 2 & -1 \\ 1 & -1 & 2 \end{bmatrix}$.Hence compute A^{-1}. Also evaluate $A^6 - 6A^5 + 9A^4 - 2A^3 - 12A^2 + 23A - 9I$	2
5	K3	CO3	Investigate, for what values of λ *and* μ do the systems of equations $x + y + z = 6, x + 2y + 3z = 10, x + 2y + \lambda z = \mu$ have (i) no solution (ii) unique solution (iii) infinite solutions?	2

Answers:

1. $A^{-1}=\begin{bmatrix}1/2 & -1/2 & 1/2\\ -4 & 3 & -1\\ 5/2 & -3/2 & 1/2\end{bmatrix}$

2. 3

3. $\lambda=2,2,8$; $X_1=k_2\begin{bmatrix}1\\2\\0\end{bmatrix}+k_3\begin{bmatrix}-1\\0\\2\end{bmatrix}$, $X_2=k_1\begin{bmatrix}2\\-1\\1\end{bmatrix}$

4. $A^{-1}=\dfrac{1}{4}\begin{bmatrix}3 & 1 & -1\\ 1 & 3 & 1\\ -1 & 1 & 3\end{bmatrix}$

$$A^6-6A^5+9A^4-2A^3-12A^2+23A-9I=5A-I=\begin{bmatrix}9 & -5 & 5\\ -5 & 9 & -5\\ 5 & -5 & 9\end{bmatrix}$$

5. $(i)\ \lambda=3\ ,\ \mu\neq 10\ \ (ii)\ \lambda\neq 3\ ,\mu$ *may have any value* $(iii)\ \lambda=3\ ,\ \mu=10$